

Transformation Seminar: Me and my Markov blanket

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hello first i'd like to warmly welcome you to the anna freud center to our second series of transformation seminars today is the third seminar of the series we are really delighted that so many of you are able to join us these debates are about your questions about your thoughts about us collectively developing a new vision for mental health we look forward to receiving your comments and questions and hope that you all will be inspired to continue these discussions after the seminar we're holding these seminars because we feel there is an urgent need for a national discussion around mental health of children young people and their families mental health has become a focus during the pandemic and we hope that once we are beyond the crisis we will be able to rebuild better for all of us also in terms of the mental health of our young people who are at the sharpest end of their current pressures the theme of the transformation seminars is thinking differently about mental health we invite speakers to respond to a simple question then ask them what is their idea to transform children's and families mental health in the uk the seminars explore areas beyond clinical interventions important theories are to ask about how we can think about do things differently individually collectively and as a society to reduce the growing needs we have as individuals in our complex and interconnected social world and tackle some of the underlying causes of mental ill health after the presentation we would like to invite you to ask questions you see at the bottom of the screen that there is a space for q and a's and if you'd like to submit your questions we'll try to get through as many of them as we can in the time that we have available finally if you are tweeting please tag us at afn ccf and use the hashtag transformation seminars today i am honored and delighted to welcome someone who is a world-renowned exceptional thinker in his field when i say his feel it truly is his because no one has the breadth and depth of understanding of the human mind quite like carl frista carl is a theoretical neuroscientist and a world authority on brain imaging and computational neuroscience he's one of the most innovative thinkers about mental health alive today he's welcome professor of imaging neuroscience at ucl he started off in psychiatry but has been an independent and

extraordinarily creative thinker applying his own original photo his revolutionary impact on studies of the brain derives from his inventive use of probability theory to analyze neural imaging data he invented a technique statistical parametric mapping or sbm that is now widely used almost universally to look for correspondences in brain activity as measured by magnetic resonance imaging and cognitive function he also invented voxel-based morphometry a sensitive method for measuring the volume of brain structures one classical well-known application demonstrated the increased volume of a region underlying spatial memory and the brains of london taxi drivers car is dynamic cars dynamic causal modelling is used to estimate how different cortical regions of the brain influence one another his current thinking about brain function is revolutionizing our ideas about perception consciousness mental disorder and the nature of experience itself the active inference framework or prediction a minimization is a relatively recent theory in mathematical neuroscience pioneered by carl and his collaborators it holds considerable promise to be the unifying theory of the brain it is used increasingly to support theoretical predictions in experimental research as well as advance the existing theory of perception and action my deep admiration for kyle reached new heights through reading his contributions to independence sage or scientific advisory group for emergencies a group of scientists were working together to provide independent scientific advice to the uk government and the property and how to minimize deaths and support britain's recovery from the curvy 19 crisis carl's awards and honours are far too numerous to mention i will just pick up a couple he's a fellow of the royal society a fellow of the royal society of biology in 2003 he received a golden brain award an international science award in the field of neuroscience given to people who made breakthroughs in research on vision and the prey in 2013 he was awarded the international weldon memorial prize which is bestowed on the person deemed to have published the most noteworthy contribution to the development of mathematical or statistical methods applied to problems in biology in the previous decade in 2016 he received the charles branch award for unparalleled breakthroughs in brain and the glass brain award for his lifetime achievement in the field of human brain mapping this evening we will hear carl discuss the emergence of selfhood and how our sense of ourselves emerges in interaction with the minds around us the seminar intriguingly entitled me and my markov blanket we are very very privileged to welcome one of the greatest thinkers in neuroscience today welcome carl thank you very much for that lovely introduction it's a true honour and pleasure uh you know to be with you uh this evening um i am going to take the challenge of thinking different differently um about mental health very literally uh to the extent that i feel i should apologize for um how

different this lecture might be from um the kind of lecture that you would anticipate um in this series i'm going to attempt to provide a first principle account of dietic interactions and speaking to selfhood uh as an emergent property of those interactions but i'm going to do so from the point of view of a physicist so i'm going to present a lecture of two halves the first i um again i apologize is is really a foundational course in the way things work the physics of self-organization and i'm going to lift from that an understanding of sentience as an emergent property of self-organization so i'm going to talk basically about the maths of the mechanics of mind and i'm going to illustrate some of the simpler uh principles that emerge from this treatment using simulations of primordial soup and the end point really is a synchronization between what's on the inside and what's on the outside of anything that exists um or self-organizes itself to a state of existence then i'm going to tell exactly the same story but using the words and the language of a clinician or a neuroscientist focusing on predictive processing and predictive coding formulations of how our brains work but the end point is exactly the same it's the demonstration that the things that matter in terms of our engagement with the world out there behind our sensorium rests upon a degree of synchronization that at the end i'm going to offer as a metaphor for alignment um and the thing that glues together you and me in any dyadic interaction so that's the the road map um over the next few minutes um i normally start the physics part of this uh talk with uh the question posed by scrolling how can the events in space and time which take place within the spatial boundary of a living organism be accounted for by physics and chemistry now clearly we're not going to be able to answer that question but what we can do is just focus on the notion of a boundary of a living organism and of course to define anything and specifically a living organism you have to define what separates the organism from not the organism me from everything else me from other and scrolling it would be the first person to acknowledge that this boundary is in and of itself a statistical process and i'm going to read that boundary um by borrowing from bayesian mechanics the notion of a markov blanket so a markov blanket for those of you who don't know is a very simple concept it it's a statistical set of states that insulates or separates internal states from out external states on the outside so graphically we can consider some little universe here where each of these circles corresponds to some state of this universe and the arrows or edges denote an influence of this state on that state and what i can do is i can identify a set of states here encoded by blue as internal states and then i can define the markov blanket or the blanket states in terms of the parents the children and the parents of the children so that affords a set of states that in effect provide all the information i need about the universe

beyond those states that will enable me to predict what will happen on the inside so in other words statistically speaking whatever is happening here is independent of what's happening here because given or conditioned upon these blanket states these blanket states hold all the information that i need to tell me about how i'm going to evolve about my dynamics i'm going to make a further move here i'm just going to split bipartition these blanket states mathematically into two sets one set which we're going to call sensory states that influence but are not influenced by the internal states and another set active states that are influenced by internal states but are not influenced by external states and with that partition in place i can now define the states of any particle or person a priest anything that exists as the internal states of that particle and their blanket that comprise the sensory and the active states and you may be asking why have i done that um i think the intuition is more readily available readily available if you think about some of your favorite systems so one of my favorite systems is the brain and one can consider the internal states of the brain as every state of synaptic activity efficacy connectivity that constitutes a particular internal state of the brain and these internal states in your activity are communicated or influence active states my actuators my autonomic nervous system that produces changes in the external states out there the physiological states of my body or the motor states of my um my uh motor plant that change the sensations the sensory states that are picked up and impinge upon my internal states so that i have this of circular causality where the internal states exchange for the external states through the markov blanket a dual uh carriageway where the influence of the in the inside on the outside is mediated by the active states and the influence of the outside on the inside is mediated by the sensory states and these arrows are what license my attribution of states to internal or external sentencery or active in the sense that they denote the conditional independences that we've uh just described that define the markov blanket but this structure can be found anywhere that you can separate something from the rest of the universe so for example a little cell will have intra internal states will correspond to the intracellular states the active states could be the actin filaments that support the surface states that push the surface states into the external milia the external states that reciprocate by changing the cell surface sensory receptors that cause changes in the internal states and so this perception action cycle at this very elemental level is still in play and rests upon this markovian partition into internal external and intervening blanket states so what i want you to do now is to bear that in mind but forget about it for the moment what we're going to do next is just to consider the physics of systems that self-organize and then what we're going to do is to put the markov blanket back in play and

see what implications it has for the relationship between what's on the inside and what's on the outside so this is a bit heavy again i apologize for the mathematics you don't need to understand the functional form of these equations but but i think they are useful just to sketch the architecture of the ideas and in fact um if you are mathematically minded that their simplicity so what i've done here is just drawn the kind of universe that i'm interested in and i'm interested in universes that exist that persist over time so mathematically that means that they have an attracting set um another word for that is that they constitute a random dynamical system with a global random attractor so more intuitively i've just sketched out two states here it could be the temperature of my body some say the concentration of oxygen if i was a physiologist and i've traced out over time the way that these states might change this could stand in for anything it could be very fast fluctuations and electrical and chemical constituents of a part of my brain unfold over milliseconds to give me uh synchronized activity in the brain it could be my heartbeat two aspects of my heart contracting every second or so it could be me getting up in the morning having a cup of coffee doing my emails looking forward to lunch it could be christmas easter some holidays birthday this structure has to exist at any scale at any spatial and temporal scale crucially embodying the fact that its existence depends upon revisiting the same states the neighborhood of the same states time and time again after a suitable period of time so these are the states x they change according some flow here with some random perturbations of fluctuations denoted by ω here now the reason i've gone to some length in just casting the mathematical basis of the arguments in terms of the flow of dynamical systems is a well-established and very ubiquitous formulation of the behavior of these kinds of systems that inherits from the fact that if i keep revisiting the same states then i can treat this attracting set of states that i keep revisiting that characterize me or things like me in terms of a probability distribution that you will find me in this state if you sample me any random time and i can write down the mathematics of the way that this probability distribution itself evolves but we've just said it doesn't change with time because we exist over so what we can do is we can solve um the equation describing the density dynamics it's called the Fokker-Planck equation you may know it or you may not as uh the Schrödinger equation or the master equation effectively all the same things the solution to that the particular solution for things that exist over time is remarkable in the following sense using a particular decomposition the Helmholtz decomposition it tells me immediately that things that exist will flow in a way that can be described as a gradient flow on the log probability of being in any particular state um i'm going to give you some intuition as to how this equation unpacks because i think it's quite

interesting but also just to illustrate its simplicity what we're basically saying is that well let's imagine that i placed a drop of oil in some water or on say a sheet of blotting paper now the kinds of systems that dissipate that we're not interested in that don't self-assemble and don't have this attracting set of characteristic states they would behave a little bit like this the ink molecules would diffuse throughout the solvent until they dispersed everywhere subject to the random molecular uh fluctuations the systems that we're talking about usually living systems behave very they appear to gather themselves up and assemble in a restricted number of states or places that they could be in other words it looks as if their states are flowing up concentration gradients resisting or offsetting the random fluctuations that will be dispersing them and that's all that this equation says it just says that any system that exists can be described in terms of its dynamics its flow its movement its evolution and that that flow can be decomposed into two components there's the part that gathers things up that resists the entropic erosion and dispersion dissipation decay dilution that is this component here that's more proportional to the amplitude of those random fluctuations γ and then there's another component that's right angles to that that circulates on the iso probability contours it's called solenoidal flow and this is denoted by the q here anything that exists can be written down in terms of its dynamics in terms of this decomposition into a gradient flow that flows up the concentration variance and this circular or solenoidal flow and you'll all be familiar with this very ubiquitous description of dynamics when you're looking at water flowing down the plug hole so the gradient flow is going downhill and the circular flow is the swirling around uh that one witnesses so that's how things that exist behave dynamically what about the markov well what is true of all the states is true of the active and the internal states so at the same time that the internal look as if they're performing a gradient flow on the log probability of the it is also the case that the active states are doing the same and i'm going to offer this as a mathematical image of perception changing our internal states and action changing our active states but in the service of looking as if we're increasing the same quantity and that quantity is this log probability of the blanket states so what does that mean well it's probably something that you're very familiar with depending upon um your commitments academically or the way that you understand optimization processes i've just listed a few things here and i uh and in the hope that you will recognize you know at least one if not if not all of them the first interpretation of this quantity here this log of the probability of my blanket states being characteristic of the kind of thing i am me or my markov blanket we can interpret this as simply an expression of value these are the states that i aspire to that i am generally found in

um mathematically we can approximate this quantity with a con with something known as a free energy or an evidence lower bound in machine learning but the perspective i'm selling in this interpretation is that all systems that exist will look as if they are performing a gradient flow on this value function and of course this is exactly the assumption the foundation of reinforcement learning optimal control theory and economics expected utility theory so that's one perspective an equivalent complementary perspective is on the negative value so if you did information theory this would be known as self information or surprisal or more simply surprise the negative log probability the improbability of finding my blanket states in this state here which means that we can construe the self-organization of internal active states in terms of a suppression of surprise and this leads to things like the infomax principle uh the minimum redundancy principle and indeed the free energy principle that's nice because the time average of is mathematically called entropy which means it looks as if or it will any system that exists will look as if it's trying to minimize its entropy so this clearly is the holy grail of synergetics that tries to understand the imperatives behind self-organization and if you're a physiologist it's just homeostasis it's just gathering your physiological states within bounds to preclude uh an entropic dispersion and wandering into non-viable or highly surprising physiological states i'm going to pursue another interpretation and which is more statistical in nature inferential in nature and in statistics and in particular bayesian statistics the probability of the blanket states given a markov blanket or given me can also be interpreted as something called model evidence or marginal likelihood bayesian model evidence so this is basically saying the states the internal active states are changing in a way that appears to be maximizing model evidence and that finds a nice home uh when unpacked in terms of the bayesian brain hypothesis evidence accumulation and as we've heard an introduction uh things like predictive coding so if it's the case that anything that exists should look as if it was doing some sort of prediction or modeling of what's beyond its markov blanket then we should be able to simulate that and show its emergence in any um any self-organizing system that possesses a markov blanket so what i'm going to do is briefly show you some simulations of a little universe that we've engineered precisely to show and or to ask the question can we convince this kind of mechanics this kind of synchronization between the inside and the outside that's what we've done is just built a little world comprised of 128 macromolecules uh the details not important the same phenomena emerge irrespective of the differential equations one uses all we need to know is that they're all loosely coupled um and we can see them bubbling away to um at the bottom of a potential well here and you may be asking well why have i done this what are we

going to do next well because we've written down the equations of motion that underwrite the flow that we've been talking about we know what influences what we know which molecules synthetic molecules influence which other synthetic molecules and we can preclude action resistance and in virtue of that we can now define the conditional dependencies that define the markov blanket so the hope was that we'll be able to find a little creature living in this synthetic primordial soup defined purely on the basis of its markov blankets the parents the children and the parents of the children defined by this adjacency matrix and here indeed is a little viral-like particle wiggling around in the middle of the synthetic soupy little cilium or tail here where the internal states have been color-coded with blue the active states that typically in these simple simulations usually underlie the um sensory states in the magenta here pushing them into the external states external media in cyan here so this simulation reveals that there is indeed a little creature living in the middle of this soup that's quite happily attaining and maintaining its visiting its attracting sets its technically non-equilibrium steady state but the question now is can we find any evidence that the internal states are somehow maximizing the evidence for some implicit model of what's going on on the outside in other words do internal states appear to infer the causes of their sensory states some people in philosophy like to call this self-evidencing soliciting evidence for your own existence um and it's relatively easy to demonstrate indeed this process is manifest um this example here is used to illustrate the simulation of synthetic event related potentials used in cognitive neuroscience and electrophysiology but just to quickly walk you through it because what i want to get to is this underlying construct called a synchronization manifold what we've done here is ask are there any mixtures of the internal states that will predict the motion of the external states so this is a little bit like asking are there any patterns of neuronal activity in the visual brain that accompany motion visual motion outside that is registered by our sensory epithelia and indeed it is easy to find this this has to exist and i'll explain why in a second and here is an illustration so what we've done here is take a mixture of motions fluxes of the external states um shown in cyan the prediction of those and the 90 confidence intervals based just upon the internal states remembering that the internal states can't see the external states all they know is what they um how the external states impress themselves on the sensory states and uh and the influence of the sensor states by the active states on the and we can see there's a remarkable correspondence between the implicit knowledge the inference predicated on the internal fluctuations um about what's going on on the outside despite the fact that there is no direct access and indeed if we gather together little excursions here and average them we can uh

simulate um um event related potentials here's an empirical example from monkey electrophysiology so you may be asking well how does this happen why why is it the case well it must be the case in the sense that for any given blanket state there must be an average internal state and an average external state which means there must exist um for any given blanket state the expected internal mode of activity and an external mode of activity and all we're seeing is fluctuations on this synchronization manifold and i'm using the word synchronization manifold just to bring to closure that you know the first half of this talk with a nod to what has been understood for centuries and in particular uh the notion of generalized synchronization or synchronization of chaos uh noted by christian huygens um many many years ago and most of you will be familiar with this is the observation that suspend two clocks from the same wall or beam and inevitably after a period of they will start to oscillate in synchrony they will become entrainment they were becoming aligned it can there is no other solution and this is basically just an illustration of this fundamental property of uh couple dynamical systems in this instance we're coupling the inside to the outside precariously through the so here's actually a drawing um by hogan's um with two clocks suspended from the same beam so from our point of view um one clock or one pendulum can be the internal states and the other can be and they are coupled through the beam uh in terms of the blankets debates comprising the active and the sensory so i like this construct because it um it illustrates the complete symmetry here so if you subscribe to this interpretation of synchronization where the inside is trying to you are trying to infer what's going on in the world then you also have to accept that the world is trying to infer what you're about and what you're doing and that's not announced a non sensible uh interpretation when it comes to things like niche construction and the like however um we are going to um try to get back to this notion of synchronization through another route but let me just first of all summarize what we've done uh with all the heavy lifting in terms of the physics what we're saying is that the existence of a particle or a person anything that exists implies a partition of a systems state systemic states into internal blanket namely sensory active and external states that are hidden behind the markov blanket and because active states change but are not changed by external states they're going to reduce the average surprise or the entropy of blanket states and this means that action will appear to maintain the structural and functional integrity of the blanket and some people like to discuss this in terms of self-assembly in chemistry or in the life sciences autophages hidden causes of sensory states by increasing their bayesian model evidence namely self-evidencing and actively influence those causes via what is often called

in my world active inference just one little twist here for philosophers um there is an argument the internal states in virtue of the existence of the synchronization manifold parameterize probabilistic beliefs about external states and one can unpack that in terms of information geometry and complement a cartesian dualism with a markovian monism but we're not telling that story today um what we oh so yes this is just a little interlude um this is um one of my post docs um went to america to work with mike levin at tufts and a few months after arriving um he and his wife had a little baby and his wife bought him uh a mark off blanket so here um keira the little baby with um franz's markov blanket here's markov blanket so if you mark off so if you if you want a mark of blanket they are available in america so we've done the statistics of life and what i promised i would do now is just rehearse exactly the same story but from the point of view of a trying to understand sentience and how we make sense of the world um so i'm going to unpack this part of the talk in terms of the basics the fundamentals of the anatomy of inference in terms of predictive coding um with a brief focus on clinical micro circuits and in particular sensory attenuation because i think that's a particularly important aspect of the physics of sentience particularly in psychiatry and indeed mental health and then conclude with an illustration of all this in action and my font hope is to get back to and what it means for models of the world where the world includes other things like me and indeed could include me um so the anatomy of inference all of this is predicated on the underlying notion that can be traced back to the students of plato i think most best articulated by helmholtz as we see it seen a moment the notion that perception rests upon a constructive act and it's a perspective on perception which is in contrast to 20th century sort of outside in filtering sensory information it rests upon an inside out constructive act where you construct predictions or hypotheses that best explain the signals and sensations of your sensory states of your sensory epithelium so you have to have something in mind as a potential explanation for your sensations before you can perceive it i think this is beautifully illustrated by the this painter famed for doing still life oils that when viewed from a different perspective give you a very different percept so if previously you saw a bowl of fruit and now you see a face the key thing is you brought that face to the table that is your explanation it's this inside out act that now enables you to find a better explanation a simple explanation for why you're perceiving or sensing this particular pattern of sensory signals so helmholtz would i think in the science and the life sciences and the physical sciences be the person who first articulated this most clearly nicely illustrated by this sentence or quote here objects are always imagined as being present in the field of vision as would have to be there in order to produce the same impression on the

nervous mechanism so again speaking to the notion that it is um the impressions you get on the sensory epithelia are explained by what would happen if this was the cause and the cause has to be imagined in a mindful way and clearly this is very much akin to richard gregory uh and other um seminal psychologists of the 20th century who formulated perception as a hypothesis testing again this notion that using sensory evidence sensations sensory states as evidence for or against a perceptual explanation or hypothesis and if confirmed then you commit to that and that is what you see um as opposed to what you are looking at these ideas have been used to great effect in a technical uh arena um particularly machine learning by people like geoffrey hinton and in a slightly more translational setting by people like uh peter diane and indeed they came up with the notion of a helmholtz machine as a machine learning or neural network formulation of this basic idea borrowing ideas from thomas bayes and in particular richard feynman's free energy or variational solution to the problem of this difficult inference problem that we all face when finding the best explanations for our sensorium um let's just go back to helmholtz producing impressions on the nervous system so this fits very nicely with this markovian blanket construct so what we're talking about here is if it is the case that to to it to be is to perform gradient flows on evidence model evidence what i am effectively saying is i will appear to try and explain the best explanation or causes for the sensory impressions or the shadows on my sensory states my sensory veil so i will be trying to discern or represent what is causing these sensory shadows here and using this the maths that i presented in the first half um one can sketch out how our brains actually do this in a relatively straightforward way um so these equations again may look complicated but you know once you get used to them they are remarkably simple and yet may explain how our brains and our neuronal message passing actually implement this fantastic act of being able to explain what's causing to produce those fantasies um that are vertically endorsed by the sensory uh this are sensory evidence that we you know we gather with our eyes and our ears and all other organs so let's just come back to this gradient flow and focus on the internal states so here's our gradient flow on this log probability under some simplifying assumptions gaussian assumptions we can write this expression in this form here which is a mixture of two things the rate of change my internal state say my neuronal activity can be broken into the solenoidal part which is in engineering known as a prediction so it's a particular one i would the states of the world in the future without any observing any new information so nothing is actually changing in terms of the underlying surprise because it's a prediction about things i haven't yet seen and then a prediction error epsilon which is the difference between what i predicted and

what i actually and this gradient flow basically is just saying we're going to minimize the sum squared error weighted by that precision π_i here so just to try and give you an intuition of what that sort of how that would work um in a sort of uh coarse-grained fashion let's imagine that we have this sensory impression here this shadow on our sensory epithelia and we had some expectation or mean explanation μ_i here that the best explanation was of a howling dog and if i had a generative model g that could generate from this expectation what i would have seen my sensory states that would have been produced if this was right then i could compare the prediction generated by the generative model with the actual sensations and elaborate and then i can simply use my prediction error to optimize or change my expectations until the prediction error has been minimized and at which point i will have a perfectly good explanation for my sensory impressions and that changing that belief updating that evidence accumulation is just this gradient flow it's just minimizing prediction error um notice that the name of the game here is to minimize prediction error or to maximize the the probability of the sensory or the blanket states under a particular model of the world that is generating the predictions we'll never actually know what caused these sensations but that's not the point if i can get through my life minimizing prediction error job done and of course that job is done by this gradient flow um up here so this provides a nice and parsimonious very simple picture um of active sentience um so we can forget about all the focal point equations and the gradient flows and we can say that we can explain everything in terms of minimizing prediction error which means that we can understand perception as the changing of internal for example brain states to supply predictions that are as close as possible to the sensations and that would be a canonical way of minimizing prediction error and we can regard that as perception changing predictions however there is another way of minimizing prediction error we could actually go and resample our sensory states our sensations to make them more like the predictions and this involves actively new sensations from the environment by looking over there or by touching this or touching that and that gives a nice explanation for changing sensations to minimize prediction error so both action and perception are simply there in the service of minimizing prediction error um so what would that look like in a little brain and i've deliberately chosen a bird brain here because i want to use birdsong uh as a little simulation at the end just to illustrate and make concrete some of the phenomena that we've been talking about imagine this is um a piece of neural anatomy from a bird brain this is the auditory thalamus of a songbird it receives some sensations so let's assume that the bird is currently listening to for something it receives these fluctuating um sonogram representations of sensory

input time frequency and in the absence of any top-down predictions there is the sensations just to become the prediction error they go up to the higher vocal center for example where they drive through this gradient flow expectations have tried to provide predictions of what's coming in until such point as the predictions match the input and the prediction error is quenched or resolved however at the same time while these first order predictions are fluctuating being driven by the ascending prediction errors they are also in receipt of top-down predictions from a higher area say area x in the in the songbirds brain uh and the same mechanics ensues that there's a difference between the activity here and the top-down prediction that elaborates a prediction and sent forward to get better top-down predictions of the first order predictions that generate um and try to suppress the first order prediction errors and you can see how you can build this hierarchy into a deep generative model uh of the sort necessary for inferring what's going to happen next at different temple scales a narrative or in this instance uh bird song this is a useful explanation for listening but notice that we've also included in the things the sensations that have to be predicted the sensations that are reporting the status of my motor plant in this instance the um the voice box of the bird here so the if i were able to if i was listening then i would uh i would not be uh predicting any sensations from for my articulatory muscles any stretch receptor signals however i can use exactly the same generative model same machinery predictive machinery to actually reproduce and sing and all i need to do is basically switch on by increasing the precision uh or of the descending predictions of what i would feel if i was actually producing this sensory input and then i would have a difference between my top down predictions and what i'm actually sensing from my muscles and i would have a prediction error and i could use that prediction error to drive my expectations and in fact i wasn't singing or i can send them back outside to the world and actually cause the muscles to move until they supply the predicted signals so all i've done here is sketch a computational anatomy that is isomorphic as a classical reflex arc where the way that we move and indeed the way that we secrete in terms of autonomic action is rests upon top down providing a set point for classical reflex arcs that these peripheral prediction errors actually drive and fulfill and all this requires is a sort of yin yang or a selective gating of whether i am attending to switching on the precision of the prediction errors pertaining to proprioception to action or extra reception in this instance auditory um sensations so this illustrates a very important role of um of precision in contextualizing whether i'm passively listening and not moving or whether i'm actually generating the sensations actively and implicitly also attenuating the consequences of my own action sometimes referred to as

sensory attenuation technically what we are the story that we've just told this little predictive coding with reflexes formulation of active inference is a story about the top-down predictions descending predictions playing the role of coral red discharge predicting stuff that i'm sensing in the extra receptive domain and at the same time bilaterally generating predictions of proprioception that are interpreted as motor commands by the peripheral nervous system and my neuromuscular junctions from the point of view of the control of precision this is often this mechanism sort of switching between sensory and sensory attenuation sensory attention and selection uh sometimes interpreted in terms of um attention being if you like the psychological homolog of this sort of precision engineered message passing here i just want to drill down on that because i'm going to use that in the final couple of slides um to simulate the attic interactions um so just to show you how potent this precision modulation of the prediction errors that we're trying to uh that we use to select the kinds of prediction errors that we want to minimize on that we believe convey precise information for our belief updating i'm going to play a little game if you allow me to play this game um it's illustrative of one of the clearest examples of um sensory attenuation which is called circadic suppression but i'll explain how and why just after illustrating its potency so this is not going to work very well given that you're all very remote but could you if you are a man do one task and if you're a woman do another task and the task is as follows in a moment sherlock holmes is going to jump from this side of the screen to the other side of the screen and your job is to tell me whether any aspect of this picture of sherlock holmes changed when it goes from here to here i'd like the women to focus on this fixation point here in the so they have to keep their eyes still and um tell me whether sherlock holmes changed on pass on whilst fixating on the cross conversely the men i'd like you to focus on shark homes and try and track sherlock holmes and what i want to know is who's the best at being able to detect whether or not anything changed uh in the um picture of sherlock holmes so if you if you can all do that i'm going to pick on um lydia as representative of women to give to give me her answer and tim as representative of men to give me uh give me um the uh given the men's answer so lydia and tim can you can you focus if you're still there on on this little particular little task right so three two one and then it's going to jump three two one lydia can you hear me hi yes i can what did you did you think sherlock holmes changed in any aspect well i saw something flat up in the middle oh what did you see i think it said did you miss me okay that's tim okay okay tim are you there i'm there i'm here yes i'm afraid i didn't see anything i didn't no you didn't see anything at all no change jonah combs so beautiful that that is a lovely illustration of the power of psychotic suppression

or sensitive attenuation um um in the oculomotor context so what lydia well done that was you only had about 50 milliseconds to see that but you did indeed see did you miss lee and and the idea here is the reason that you lydia saw this um was that you were not suppressing the precision of your visual prediction because you weren't moving tim on the other hand had to suppress the prediction the visual prediction errors because they were a consequence of its active sensing or visual palpation of the world so this is a lovely example of enforced sensory attenuation where we're basically transiently blind or transiently suspend our attention when we cause the sensations and of course when we move our eyes there is an enormous visual flow or optic flow induced simply by our eye movements which we cannot see and in virtue of the fact that tim could not see it whilst he was moving his eyes he couldn't see anything else either which is why you lydia were the only person that could actually see the digi did you miss me so i think this is a wonderful example of you know hardwired sensory attenuation that really has a profound influence of on on how the sensorium updates our belief and how that depends exquisitely on agency and action and what i want to do now is just to close by leveraging um sensory attenuation and augmentation that underwrites whether i'm whether i'm listening or talking or whether i'm um listening or singing to simulate now um daddick interactions so previously we've been talking and simulating little creatures um into exchanging with the external world what i'm doing now was making a further and putting two markov blankets um face to face as it were so the internal states of one blanket are the external states of another in the hope of simulating dyadic interactions but to do that to preserve the conditional independences i have to do turn taking and i'm going to use this precision mechanism to turn take so one in this instance is going to be a bird is and then it's singing and then it's listening and then it's singing and vice versa for the other bird so when we um integrate those equations and just let the birds sing and when we integrate those equations of motion that predictive coding equation with the prediction and uh gradient uh the prediction of minimization part of a neuro of a neurobiologically um plausible sword for the generative model that i've just described for singing and listening we can simulate what would happen if two birds were singing in the same locale what i've done here is separate the two birds so that they can't hear each other so what you're listening to now is what one bird is actually hearing it can't hear itself so it can hear itself but it can't hear the other bird and they're now simply going to repeat the simulation but bringing the two birds within hearing range of each other so now they can hear each other and then i'm going to track the internal states at different levels in this synthetic cortical hierarchy that are providing the predictions and the prediction errors

for the song that you've just heard and now if we listen to what is perceived by the first bird you can now hear a continuous melody and this melody is composed of itself singing and then hearing the other bird then itself and the other bird and in contrast to the previous slide what we're seeing here is a and this synchronization ensues because both birds have the same generative model they both have the same beliefs about what's out there and in this instance what's out there is as itself which means that this is a really powerful way to minimize surprise or prediction error in the sense that if you are like me and we are in exchange then you become very predictable from my point of view because anything that i do you are doing and vice versa and this holds for you as well so dadically altogether we can now minimize our free energy or minimize our prediction error simply because where we are committing to the same narrative to the same sensory states and active that generate the sensorium and our sensorium is shared and it's mutually predictable simply because i can assume that you are like and under that assumption and when two things are sufficiently similar and they can learn mutually to be sufficiently similar then we see the emergence again of the synchronization manifold so here in a different context but it's mathematically exactly the same construct that we um that we looked at with the simulations the primordial soup what i've done is trapped out the new brain activity if you like the neural activity generating when the two birds couldn't hear each other at the different levels in this synthetic cortical hierarchy and then neural activity patterns when they can hear each other just to show you how they align how they become synchronized of course we now know that they it couldn't be in any it couldn't be in any other way um in the sense that these uh this is the kind of coupling that is the non-equilibrium steady state or it this attracting set of states that must emerge very much like huygens clocks so that's an illustration of synchronization in terms of communication and this is a very elemental communication literally singing from the same hints sheet in order to be able to infer a very elemental if you like theory of mind the the last slide is really an invitation to think about what that means for the way that we learn our genitive models of the world to exploit this free energy minima this this prediction error solution in a world that is populated constitute constitutes creatures like me i can assume that every creature like me is and we're all seeing from the same hem sheet but to get there i have to learn that the world out there comprises creatures like me so here is the challenge for a newborn first of all the newborn has to distinguish between the things that it can act upon and the things that he has no control over in order just to develop a sense a generative model that enables it to recognize those reflexes and parts of the world can be controlled and can't be controlled but the even greater challenge then is to recognize and to

learn by this surprise minimizing free energy mechanism applied to the connections and the structure and the parameters of the to learn that the best explanation for its sensorium is a fact that many of the sensations are supplied by something else another and in particular a mother that is something like me which would be the final uh the final stage a very elemental um sort of um simplistic sense of learning that there are things like me out there so having developed a sense of mother then one might suppose that that a neonate might test then the hypothesis or perhaps i am like mother and thereby leading to a sense of self so in a rather cursory way i've managed to get to the um what was promised in the introduction um from peter about agency and a sense of self but in a way really which is more an invitation um to interpret and discuss uh whether the underlying physics of sentience really makes sense in terms of speaking to these fundamental questions um about selfhood relationships with others um i'll end with a quote from fireman it's usually a technical quote but i thought this would be more appropriate in the given in the current context the highest forms of understanding we can achieve our laughter and human compassion and with that it just remains for me to thank um all the people whose ideas i've been talking about and most importantly thank you for your attention thank you very much indeed and i'm going to pass now back to peter thank you thank you so much carl i've been monitoring some of the comments in between and if i could just cite to you a few enlightening lecture thank you this is so interesting and brilliantly explained so i've got some questions but just uh to say um in kind of try and do a little bit of translation what is remarkable about your um view is really that it's it's like the picture that you should you've turned uh perception uh upside down in the kind of active inference framework um that there is um that the traditional notion of uh uh and action and everything between is not a kind of the end point of a whole long line of processing but really something that we generate from [Music] and then compare without there now those of us here working clinically and a number of questions from that perspective have known this all along uh we have you're not telling us anything new you've known this that that people just interpret their world and you know they try and minimize prediction error and they change the world until the prediction error is minimized and some of that uh has very painful consequences in some ways it has distressing consequences in psychopathology and one of the questions and there'll be more technical questions one of the questions that people really want you to address just a little bit because i know that you have done uh is how would you brilliant though your your model is um how how would you um see uh mental disorder how would you see psychopathologies in terms of prediction error and does it is there a risk uh which i don't think there is but but i think we

could do with the reassurance that is there a risk that some sort of innate disruption in uh minimizing prediction errors could be at the root of mental disorder or is that a consequence of historical experience in terms of a consequence of uh personal experience over to you carl excellent question um so just speaking to this sort of you know turning things on their head it is i think a nice example of what uh dan denick calls a strange inversion but clearly not strange to you and people actually work with with with people um but for some um people in machine learning and artificial intelligence it's certainly a strange inversion but much more natural for people to work with actually also real artifacts of artificial sort in robotics but the the deeper i think more challenging question which i'm sure we'll drill down to is you know where does this mechanics um yield some interesting um hypotheses false inference and the uh competent behaviors that attend or belief states that attend false inference that we see psychopathology of its various flavors and i think you've you you've carved up the answer into two domains already which is part of it is this false inference in the moment what leads what's what's the um the first principle account of a false belief or indeed a false percept um and um in the moment and then there's the other um questions which are more about the learning of the generative models so there's the deployment of a generative model inverted in the moment given the sensory evidence given all the evidence from your interception from your pro-social at hand to use your genitive model to infer what's going on at the moment and you know how i can sample or create new information to make my predictions come true as opposed to in a developmental or indeed actually in any experience dependent setting the um the updating of the parameters the contingencies the lawful aspects the structure of that genetic model that would underwrite you know the emergence of a sense of agency attachment uh possibly a description of you know the um development of a daddy relationship over over you know days months weeks and years so in in the first domain um there's a really practical answer and it just comes from the empiricism um that um um you know one can find in academic psychiatry and academic neurology when one looks at the the process theories and neuronal process theories that mediate this gradient flow then it seems that the both the um the nature of false inference the psychopathology if you and the neurophysiology almost universally identifies a failure of that precision so it's a failure of that sensory attenuation that we were illustrating with the um psychedelic suppression but at a much higher level so whether you're talking about parkinson's disease failure to um attenuate proprioceptive information that i'm not moving i'm therefore never being able to develop the fantasy that i'm moving leading to psycho psychomotor poverty or whether it's the hallucinosis of organic psychosyndromes or

indeed schizophrenia where you um or perhaps the poster child here would be autism um where you are unable to modulate or attenuate the precision afforded sensory evidence so you become glued to the sensorium you fail to develop deep centrally coherent genitive models of your world you cannot disattend nothing is surprising because nothing uh ev everything is is worthwhile predicting so you can't have illusions you can't have mismatched negativities or ball responses all cardinal signs of you know of things like schizophrenia so you put that failure of sensory attenuation in a developmental context and just ask what it must be like to be born into a world that you're trying to make sense of you know and all your sensing is possibly a rattle or suckling uh or nurturing behavior or fillative touch from your mother and you're trying to make sense of the world but you can't disattend from anything to actually drill down and start to build a deep model that provides a passiveness explanation of the kind ah mother is a creature she is an object out there that's doing stuff to me i cause this bit and she calls that mid so you know then the game gets exquisitely difficult um in terms of understanding the profound implications of a failure to suspend your attention in a in an apt way to realize your agency or disambiguate i cause that or you caused there is a finding that um originally uh john watson brought forward to the developmental world that babies up to the age of about three months uh and attend uh with preference things that are perfectly contingent with their actions after three months they prefer to attend to things that are clearly but uh clearly not perfectly contingent nearly contingent but not perfectly contingent and that autistic children do not have this switch at the three-month switch would this be consistent with what you're suggesting yes it would i didn't know that's fascinating i mean you know if i if i had to build um a baby in silicon that's exactly what one would hope would happen in the normal development you've got to learn what bits of this world um you know can i act upon and what bits can't you know that's that that's the first thing that you need to do to just the self-discovery without any sense of selfhood per se but then you get to the more interesting bits well you know i thought i could control this but perhaps there's another agents and then you start to you know get it find that much more epistemically uh attractive it has much more epistemic affordance these things i thought i could control but in fact someone else might be out there let me try and uh squeeze another question so enlightening lecture thank you i'm a psychologist working with parents who struggle to relate to their kids who engage in very high risk behaviors part of the issue is that often parents overreact emotionally feeding into the behavior staying calm and acting counteractively is a challenge so thinking about the mark of blanket model where would you start with these parents not an easy question it's no because i i

was immediately trying to answer it as a psychiatrist you're a physicist as a mathematician um as a mathematician all this very simple-minded um treatment of dyadic interactions brings to the table is the um the notion of generalized um synchronization between you and your loved ones or you and your clients or you and your and your patients which would basically says that to have a harmonious aligned interaction you have to be in a like mind and singing from the same hymn sheet so if you are at all um that him sheep covers all modalities from interception from verbal non-verbal cues but the synchronization the alignment is still in play even to the level you can measure it in terms of um your body movements blushing for example pupillary dilatation all of these subliminal cues that tell you about the state of your embodied brain are mutually available um and the trick i would imagine to try and synchronize them so um just on that mathematical basis uh you have to put yourself in the same state of mind as the child in the font hope that the child will um reciprocate by putting themselves in the same state of mind as you one little trick here is that the weight the weight at which the weight the rate at which two couple systems converge to their common ground depends upon the precision or the confidence in the predictions of the two so if there's an asymmetry there you can often get a teacher or a parent or you're a therapist who has more if you like precise beliefs higher in their their model that will gent that will move towards you know the the the less precise um markov blanket or or client or child but will move less quickly than the child moves to the parent um and clearly you know you see that in a natural setting just in terms of schooling you know it's generally the case that the teacher is older than the child and the teacher is more confident about what the child knows uh but at some point you know there is a meeting in the middle but it's not always in the middle so that was a really stupid answer but at least it was faithful to the mathematics uh one final question um to to test uh that breadth that i was posting on on your behalf and here is sorry could a problem in the stability or structuring in the marco blanc can be thought of as a failure or weakness of cyclic defense and would the introduction of entropy or dissolution point to links with freud's ideas of the deaf instinct yes there are all sorts of what really interesting ways you could take that question in terms of structure learning um you know but let me focus on you know on the freudian um the freudian uh interpretation or homologs here um i think you know you if there is part of the audience that likes this um to understand the parallels and the homologous and isomorphisms between your freudian take um on free energy and the the more mathematical richard feynman take on it then mark soames is your man you know he's you know he's written beautifully about this and um would normally um start the story by noting that um

freudian formulations um inherited quite a lot from helen holtz so you know had uh had freud known fireman and had you know tried to develop helmholtz in ideas it's more than likely his free energy would be iso the same as that used by people like jeffrey hinton richard feynman um and the interestingly when when mark zones goes into this in great detail he actually um converges on the same thing that we were talking about before which is this um um that the the you know i'm not sure that we can in this we don't have time to get to ego dissolution and the like um but important um auto poetic aspect that maintains the integrity of the markov blanket is exactly this attending and disattending it's selective and appropriate switching on and off of how the sensory states get in and influence the internal states and then vicariously your actions upon the world if that breaks everything falls to pieces um and that certainly is mark seaman's story um based upon sort of you know um a sort of a theoretical neuropsychology based upon um the importance of ascending modulated neurotransmission systems um for example dopamine and the but not just dopamine all these classical neuromodulators because if it's the physiological um mediation of this precision control is via selecting selection and per synaptic gain which is neuromodulation so it's not surprising that a lot of the neurochemistry of and the pharmacology of psychopathology fixes upon these neuromodulators and furthermore this is exactly mark soda's conclusion in terms of the um the failure uh to resolve free energy um uh lives you know the seat of the soul is essentially the source of origin um of the ascending modulated neurotransmitter systems that are exactly there to provide this gating attenuation selection uh behavior um that um that preserves the markov blanket in your in the sense that we've been discussing this is an absolutely brilliant lecture uh you are everything that i have advertised you to be it is uh obviously a revelation but a revelation in the direction of helping us understand sociality helping us understand how ultimately the brain and our capacity to uh create a uh create a world that's consistent with our hypothesis and actually uh falls into uh a a continuum that uh we are living every day uh in in but as you said uh in a very strange uh as a very strange function of time uh it's a very strange function of time and uh it's uh uh to have a mind like yours that actually can bridge uh the uh obstructs writings of psychoanalysts uh with uh the cutting-edge writings of uh physicists and the neuroscientists in between and whilst maintaining an identity uh as a somebody who's concerned with mental health you're a remarkable human being you really are and we are enormously privileged to have had you here this evening and to be able to benefit from your distance so thank you so much for joining us thank you thank you thank you so much